

London equations and Meissner effect

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1 London equations

The London equations are fundamental in the study of superconductivity, describing how superconductors respond to electric and magnetic fields. They were proposed by the London brothers, Fritz and Heinz, in 1935. There are two London equations, and I'll derive them step-by-step based on physical principles, assuming a superconductor's behavior involves a perfect conductor with a specific current response.

First London Equation

The first London equation relates the electric field to the time derivative of the supercurrent. Start with the idea that in a superconductor, charge carriers (Cooper pairs) move without resistance. Assume these carriers have a charge $q = -e$ (for electrons), effective mass m^* , and number density n_s . The supercurrent density $\mathbf{J}_s$ is given by:

$$\mathbf{J}_s = n_s q \mathbf{v}$$

where $\mathbf{v}$ is the velocity of the superconducting electrons.

In the presence of an electric field $\mathbf{E}$, the equation of motion for these carriers, without any scattering (since resistance is zero), is Newton's second law:

$$m^* \frac{d\mathbf{v}}{dt} = q \mathbf{E}$$

Substitute $q = -e^*$ (since the carriers are electrons):

$$m^* \frac{d\mathbf{v}}{dt} = -e^* \mathbf{E}$$

Now, take the time derivative of the current density $\mathbf{J}_s = n_s(-e^*)\mathbf{v} = -n_s e^* \mathbf{v}$:

$$\frac{d\mathbf{J}_s}{dt} = \frac{d}{dt}(-n_s e^* \mathbf{v}) = -n_s e^* \frac{d\mathbf{v}}{dt}$$

Substitute $\frac{d\mathbf{v}}{dt} = -\frac{e^*}{m^*} \mathbf{E}$ from the equation of motion:

$$\frac{d\mathbf{J}_s}{dt} = -n_s e^* \left(-\frac{e^*}{m^*} \mathbf{E} \right) = \frac{n_s e^{*2}}{m^*} \mathbf{E}$$

This is the first London equation in differential form:

$$\frac{d\mathbf{J}_s}{dt} = \frac{n_s e^{*2}}{m^*} \mathbf{E}$$

It shows that the supercurrent accelerates proportionally to the electric field, consistent with zero resistance.

Second London Equation

The second London equation relates the magnetic field to the supercurrent, explaining the Meissner effect (expulsion of magnetic fields from a superconductor). Start with the magnetic field $\mathbf{B}$ and its relation to the vector potential $\mathbf{A}$:

$$\mathbf{B} = \nabla \times \mathbf{A}$$

In a superconductor, the London brothers proposed that the supercurrent responds to the magnetic field in a way that ensures field expulsion. Consider the canonical momentum for a charged particle in a magnetic field, $\mathbf{p} = m\mathbf{v} + q\mathbf{A}$. For superconducting electrons, they hypothesized that the macroscopic wavefunction's rigidity implies the velocity is tied to the vector potential. They assumed:

$$m^*\mathbf{v} + q\mathbf{A} = 0$$

With $q = -e^*$:

$$\begin{aligned} m^*\mathbf{v} - e^*\mathbf{A} &= 0 \quad \Rightarrow \quad m^*\mathbf{v} = e^*\mathbf{A} \\ \mathbf{v} &= \frac{e^*}{m^*} \mathbf{A} \end{aligned}$$

Substitute into the current density $\mathbf{J}_s = n_s q \mathbf{v} = -n_s e^* \mathbf{v}$:

$$\mathbf{J}_s = -n_s e^* \left(\frac{e^*}{m^*} \mathbf{A} \right) = -\frac{n_s e^{*2}}{m^*} \mathbf{A}$$

This is the second London equation:

$$\mathbf{J}_s = -\frac{n_s e^{*2}}{m^*} \mathbf{A}$$

To connect this to the magnetic field, take the curl of both sides:

$$\nabla \times \mathbf{J}_s = \nabla \times \left(-\frac{n_s e^{*2}}{m^*} \mathbf{A} \right) = -\frac{n_s e^{*2}}{m^*} (\nabla \times \mathbf{A})$$

Since $\mathbf{B} = \nabla \times \mathbf{A}$:

$$\nabla \times \mathbf{J}_s = -\frac{n_s e^{*2}}{m^*} \mathbf{B}$$

Now, use Maxwell's equation $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}_s$ (assuming no displacement current, as in the static case):

$$\nabla \times (\nabla \times \mathbf{B}) = \nabla \times (\mu_0 \mathbf{J}_s) = \mu_0 (\nabla \times \mathbf{J}_s)$$

Substitute $\nabla \times \mathbf{J}_s = -\frac{n_s e^{*2}}{m^*} \mathbf{B}$:

$$\nabla \times (\nabla \times \mathbf{B}) = \mu_0 \left(-\frac{n_s e^{*2}}{m^*} \mathbf{B} \right)$$

Using the vector identity $\nabla \times (\nabla \times \mathbf{B}) = \nabla(\nabla \cdot \mathbf{B}) - \nabla^2 \mathbf{B}$, and since $\nabla \cdot \mathbf{B} = 0$:

$$-\nabla^2 \mathbf{B} = -\frac{\mu_0 n_s e^{*2}}{m^*} \mathbf{B}$$

$$\nabla^2 \mathbf{B} = \frac{\mu_0 n_s e^{*2}}{m^*} \mathbf{B}$$

This is a Helmholtz equation, showing that $\mathbf{B}$ decays exponentially inside a superconductor, with a penetration depth $\lambda = \sqrt{\frac{m^*}{\mu_0 n_s e^{*2}}}$, confirming the Meissner effect.

Final Forms

First London Equation:

$$\frac{d\mathbf{J}_s}{dt} = \frac{n_s e^{*2}}{m^*} \mathbf{E}$$

Second London Equation:

$$\mathbf{J}_s = -\frac{n_s e^{*2}}{m^*} \mathbf{A}$$

These equations together describe the electrodynamic behavior of superconductors, capturing both their perfect conductivity and magnetic field expulsion.

Solving the Differential Equation

This is a Helmholtz equation, indicating that the magnetic field $\mathbf{B}$ decays exponentially inside the superconductor. Consider a semi-infinite superconductor occupying $x > 0$, with a magnetic field $\mathbf{B} = B_0 \hat{z}$ applied parallel to the surface at $x = 0$. The equation becomes one-dimensional:

$$\frac{d^2 B_z}{dx^2} = \frac{1}{\lambda_L^2} B_z.$$

The general solution is:

$$B_z(x) = A e^{-x/\lambda_L} + C e^{x/\lambda_L}.$$

Since $\mathbf{B}$ must remain finite as $x \rightarrow \infty$, $C = 0$. At the boundary $x = 0$, $B_z(0) = B_0$, so $A = B_0$. Thus:

$$B_z(x) = B_0 e^{-x/\lambda_L}.$$

Interpretation – The Meissner Effect

The solution shows that the magnetic field penetrates only to a depth λ_L (typically on the order of nanometers) and is exponentially attenuated inside the superconductor. For $x \gg \lambda_L$, $\mathbf{B} \approx 0$, meaning the magnetic field is expelled from the bulk of the material. This expulsion is the Meissner effect: a superconductor behaves as a perfect diamagnet, with magnetization $\mathbf{M} = -\mathbf{H}$ (where $\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$), leading to $\mathbf{B} = 0$ inside.

2 Conclusion

The Meissner effect emerges from the London equations combined with Maxwell's equations, demonstrating that superconductors exclude magnetic fields from their interior, a hallmark of perfect diamagnetism. This derivation assumes a type-I superconductor in the absence of external currents or vortices, consistent with the original experimental observations.